

SPL Front-End User Manual Project group: 27 Group Members Full name Student number Zaman Bassa u24744931 Unathi Tshakalisa u2473081 Zelamene u24735672 Mosa u24742083 Purpose This program performs lexical analysis followed by syntax analysis for an SPL source file. On success it writes the generated syntax tree as XML. Windows Usage

1. Place group-27.exe in a writable folder.
2. Open PowerShell in that folder. 3. Run the executable with an input SPL file and an output XML file:

.oup-27.exe pathorogram.spl
pathoree.xml The first argument is the input file. The second argument is the XML output path. Parent folders for the output file are created by the program when needed. When no arguments are supplied, the program reads inputple.spl and writes outputtree.xml relative to the current folder. Output and Errors Successful execution prints the source, token list, lexical-analysis result, syntax-analysis result, and the XML output location. A lexical or syntax error is printed to the error stream and the program exits with a non-zero status. The SPL lexer requires a blank separator after each token. Spaces, carriage returns, and line feeds are accepted as separators. The dollar sign is not part of an input file; it is only the parser end-of-input notation used in the specification. Example For an input file containing: : : print "hello" ; run:

.oup-27.exe example.spl example-tree.xml A successful run creates example-tree.xml. Requirements The submitted Windows executable is self-contained and does not require a separate Java installation. The executable must be run with access to the input file and permission to write the requested XML output.